# A PROJECT REPORT ON LOAN APPROVAL PREDICTION SYSTEM USING MACHINE LEARNING

Submitted in Partial Fulfillment of the Requirements for the Completion of the Summer/Industrial Internship Program at MNNIT

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# CHAPTER 1: INTRODUCTION

## 1.1Background

In today's banking and financial sector, loan approval is one of the most important decision-making processes. Financial institutions receive a large number of loan applications daily. Evaluating these applications manually requires significant time and effort and may result in inconsistencies.

Machine Learning provides a data-driven approach to automate loan approval decisions. By analyzing applicant information such as income, education level, credit history, marital status, and property area, machine learning models can predict whether a loan application is likely to be approved or rejected.

The use of predictive analytics in loan approval can improve efficiency, reduce processing time, and support fair decision-making.

## 1.2Problem Statement

The objective of this project is to develop a Loan Approval Prediction System that can predict whether a loan application should be approved based on applicant details. The system should assist financial institutions in making faster and more accurate loan approval decisions.

## 1.3Objectives

The main objectives of this project are:

- To study and understand the Loan Prediction dataset.
- To perform data preprocessing and feature engineering.
- To analyze the dataset using Exploratory Data Analysis (EDA).
- To build classification models for loan approval prediction.
- To evaluate model performance using appropriate metrics.
- To deploy the final machine learning model using Streamlit.
- To provide real-time loan approval predictions through a user-friendly interface.

# CHAPTER 2: LITERATURE REVIEW

## 2.1Existing Approaches

Loan approval prediction has been widely studied in the field of Machine Learning and Data Analytics. Traditionally, banks and financial institutions relied on manual verification and rule-based systems to evaluate loan applications. These methods required significant human effort and were often time-consuming.

With the advancement of Machine Learning, predictive models are increasingly being used to automate the loan approval process. Various classification algorithms such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, and XGBoost have been applied to predict loan approval status based on applicant information.

Machine Learning models analyze historical loan data and identify patterns that influence loan approval decisions. These systems help reduce processing time and improve consistency in decision-making.

## 2.2Research Studies

Several research studies have explored the use of Machine Learning techniques in loan approval prediction.

Researchers have found that applicant income, credit history, loan amount, education level, and employment status are among the most influential factors affecting loan approval decisions.

Studies comparing different classification algorithms have shown that Logistic Regression provides reliable and interpretable results for binary classification problems. Random Forest and ensemble-based techniques often improve prediction accuracy by combining multiple decision trees.

Recent research emphasizes the importance of data preprocessing, feature engineering, and proper model evaluation in improving prediction performance. The integration of machine learning into financial services has enabled organizations to make faster, data-driven decisions while minimizing human bias.

## 2.3Summary

The literature review indicates that Machine Learning is an effective approach for loan approval prediction. Various classification algorithms have been successfully applied to similar problems. Based on previous studies and the nature of the dataset, Logistic Regression and Random Forest were selected for implementation in this project.

# CHAPTER 3: METHODOLOGY

## 3.1Dataset Description

The dataset used in this project is the Loan Prediction Dataset obtained from Kaggle. The dataset contains information related to loan applicants and their loan approval status. The dataset includes demographic, financial, and credit-related information that influences loan approval decisions.

The major attributes present in the dataset are:

•Gender
•Married
•Dependents
•Education
•Self_Employed
•ApplicantIncome
•CoapplicantIncome
•LoanAmount
•Loan_Amount_Term
•Credit_History

•Property_Area

Target Variable:

•Loan_Status

The target variable indicates whether the loan application is approved or rejected.

## 3.2Data Preprocessing

Data preprocessing is an important step in machine learning. Raw data often contains missing values, inconsistencies, and categorical variables that need to be processed before model training.

The following preprocessing steps were performed:

- Removed unnecessary columns such as Loan_ID.
- Checked the dataset structure and data types.
- Handled missing values using appropriate techniques.
- Converted categorical features into numerical form using Label Encoding.
- Verified data quality before model training.

These preprocessing steps improved the quality of the dataset and prepared it for machine learning algorithms.

## 3.3 Feature Engineering

Feature engineering was performed to create meaningful features that improve model performance.

A new feature called TotalIncome was created using the formula:

TotalIncome = ApplicantIncome + CoapplicantIncome

This feature represents the total income available to the applicant and provides a better indication of financial capability.

## 3.4 Train-Test Split

The processed dataset was divided into training and testing datasets.

- Training Data: 80%
- Testing Data: 20%

The training dataset was used to train machine learning models, while the testing dataset was used to evaluate their performance on unseen data.

## 3.5 Algorithms Used

The following machine learning algorithms were used in this project:

### 3.5.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. It predicts the probability of loan approval based on applicant information.

Advantages of Logistic Regression:

- Simple and easy to interpret
- Efficient for binary classification
- Fast training and prediction

### 3.5.2 Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to generate predictions.

Advantages of Random Forest:

- Handles complex relationships in data
- Reduces overfitting
- Provides robust predictions

## 3.6Methodology Flow

The complete workflow followed in this project is:

Dataset Collection

Data Preprocessing

Exploratory Data Analysis (EDA)

Feature Engineering

Model Training

Model Evaluation

Model Selection

Streamlit Deployment

This methodology ensured the systematic development of an accurate and reliable Loan Approval Prediction System.

# CHAPTER 4: IMPLEMENTATION

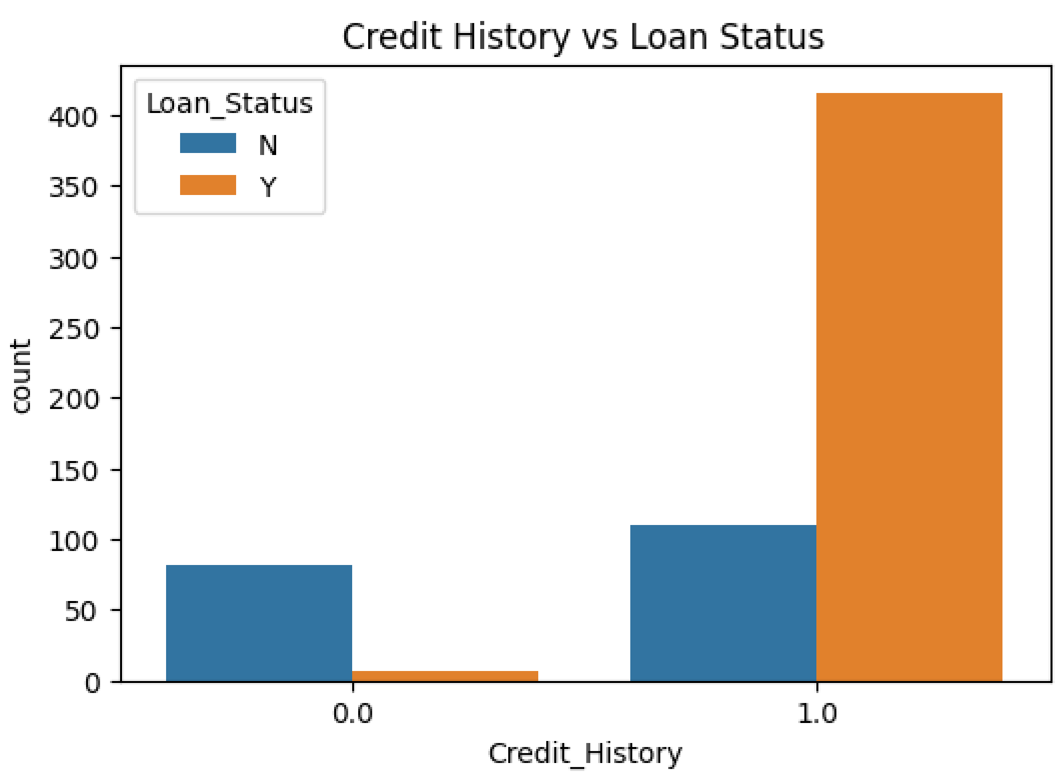
## 4.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the dataset and identify important patterns and relationships among variables. EDA helps in understanding the characteristics of the data before model development.

The following visualizations were created:

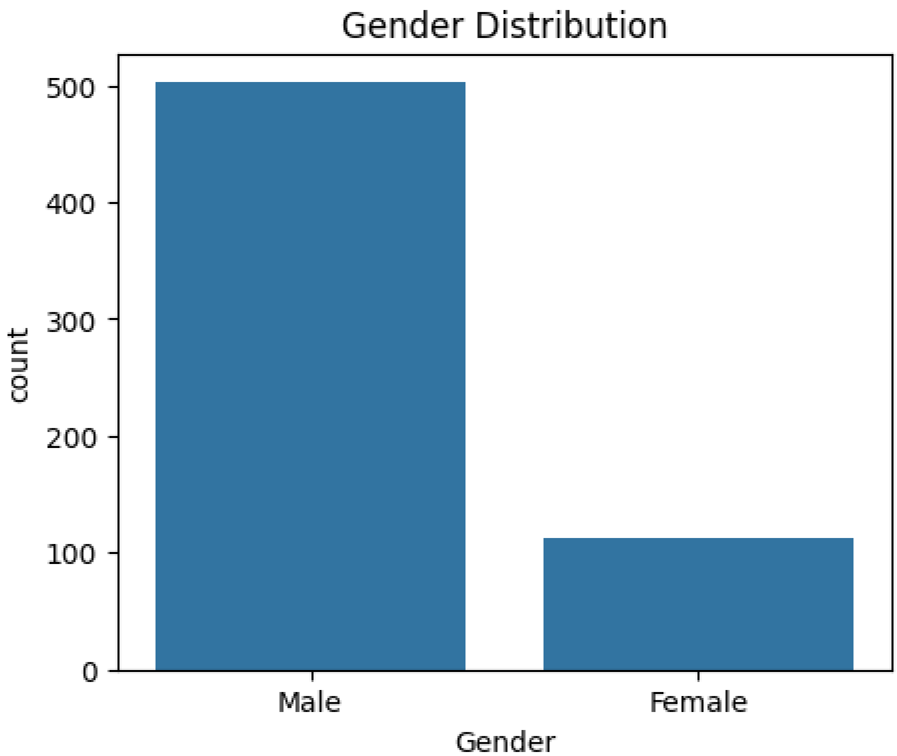
### 4.1.1 Loan Status Distribution

A count plot was generated to analyze the distribution of approved and rejected loan applications. This helped in understanding the balance of the target variable.



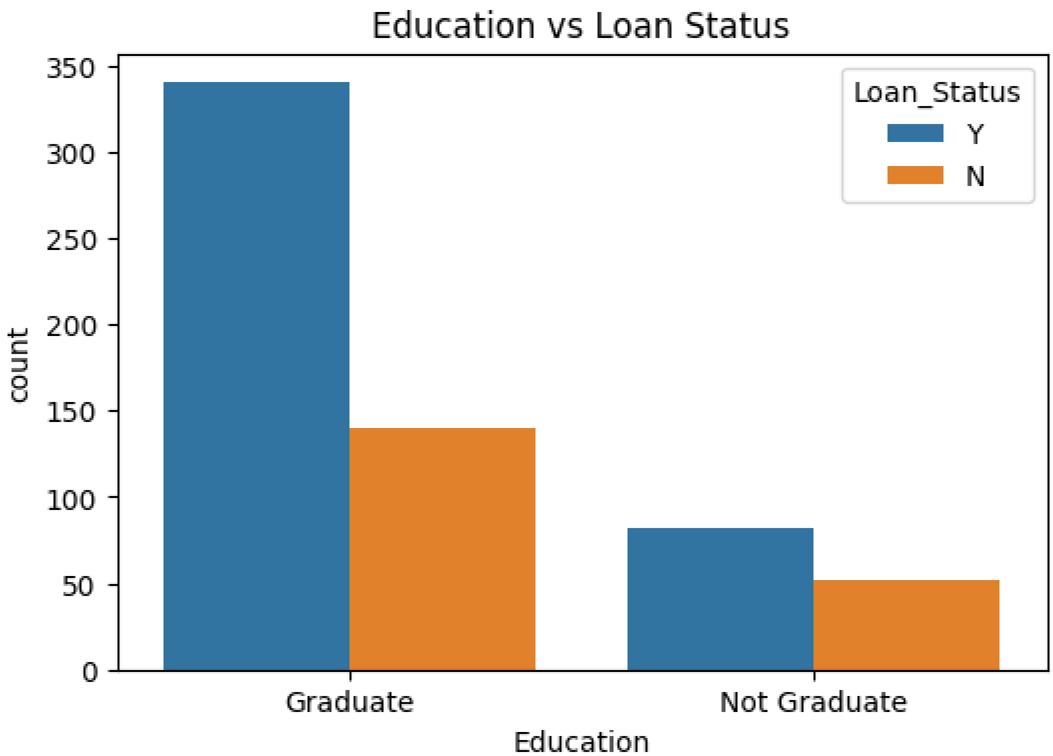

### 4.1.2 Gender Distribution

The distribution of applicants based on gender was visualized to understand demographic representation within the dataset.



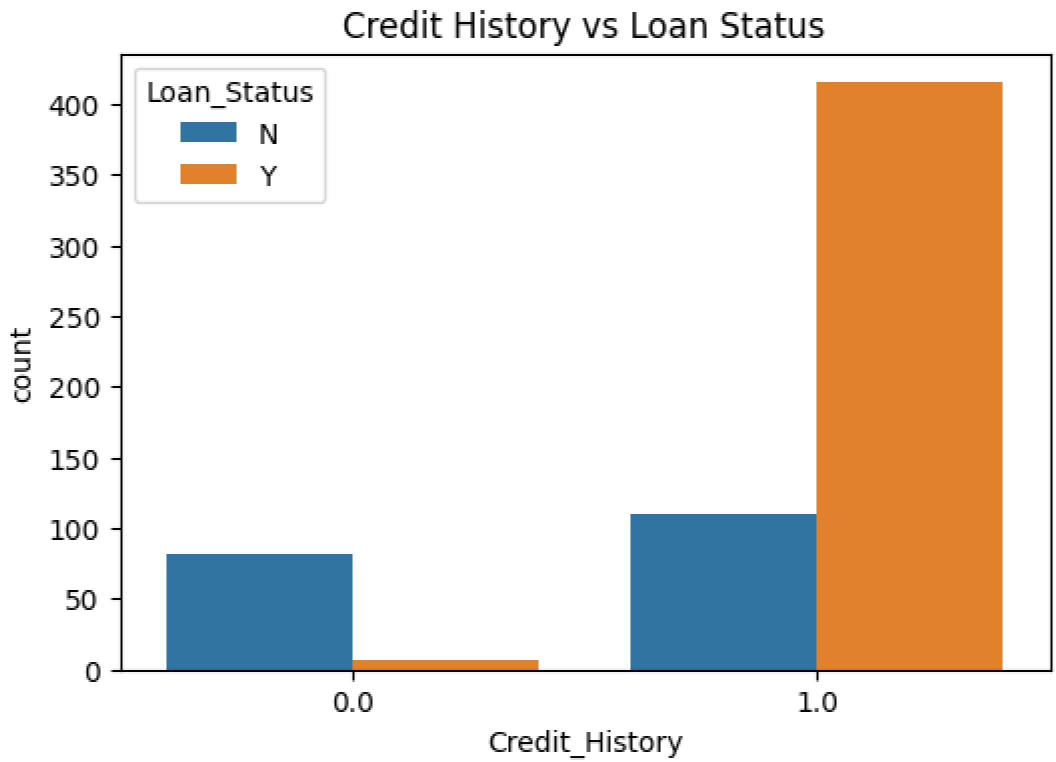

### 4.1.3 Education vs Loan Status

A comparison between education level and loan approval status was performed. This analysis helped identify the impact of education on loan approval decisions.



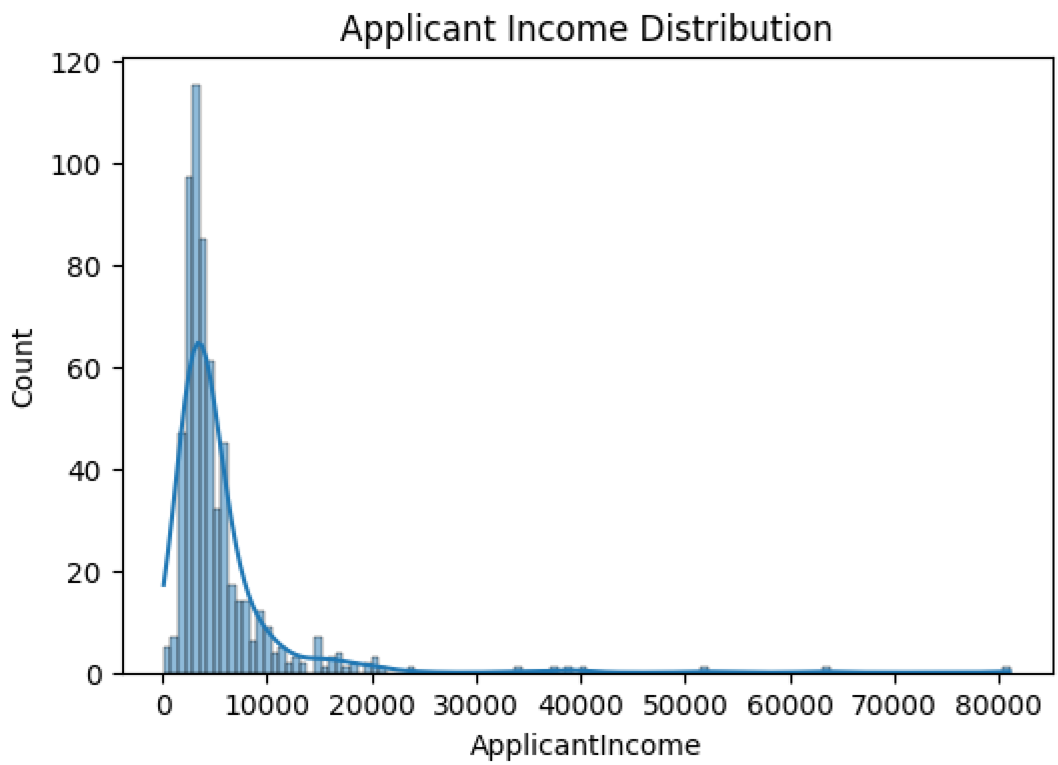

### 4.1.4 Credit History vs Loan Status

Credit history is one of the most important factors influencing loan approval. A visualization was created to study the relationship between credit history and loan approval status.




### 4.1.5 Applicant Income Distribution

A histogram was used to analyze the distribution of applicant income and understand income variation among applicants




## 4.2Feature Engineering

Feature engineering was performed to improve model performance and extract meaningful information from the existing dataset.

A new feature called TotalIncome was created using the following formula: TotalIncome =

ApplicantIncome + CoapplicantIncome

This feature represents the overall income available to the applicant and provides a better estimate of financial capability.

The following steps were also performed:

- Selection of important features
- Removal of unnecessary features
- Transformation of categorical variables

## 4.3Model Development

After preprocessing and feature engineering, machine learning models were developed for loan approval prediction.

The dataset was divided into training and testing datasets using an 80:20 ratio.

Two classification models were trained:

### 4.3.1 Logistic Regression

Logistic Regression was used as the primary classification model because it is suitable for binary classification problems.

The model was trained using the training dataset and tested on unseen data.

### 4.3.2 Random Forest Classifier

Random Forest Classifier was trained to compare its performance with Logistic Regression. The

algorithm combines multiple decision trees and generates predictions using majority voting.

## 4.4 Model Selection

Both models were evaluated using classification metrics. Logistic Regression achieved better performance and was selected as the final model for deployment.

## 4.5 Model Saving

The selected model was saved using the Pickle library. Saved

Model File:

loan_model.pkl

This file was later used in the Streamlit application for real-time predictions.

## 4.6 Streamlit Application Development

A web application was developed using Streamlit to provide an interactive interface for users. The

application allows users to:

- Enter applicant information
- Submit loan application details
- Obtain real-time loan approval predictions

The deployed application successfully demonstrates the practical implementation of Machine Learning in loan approval prediction.

A user-friendly Streamlit web application was developed to collect applicant details and provide real-time loan approval predictions.

Figure 4.7: Streamlit Application Interface !

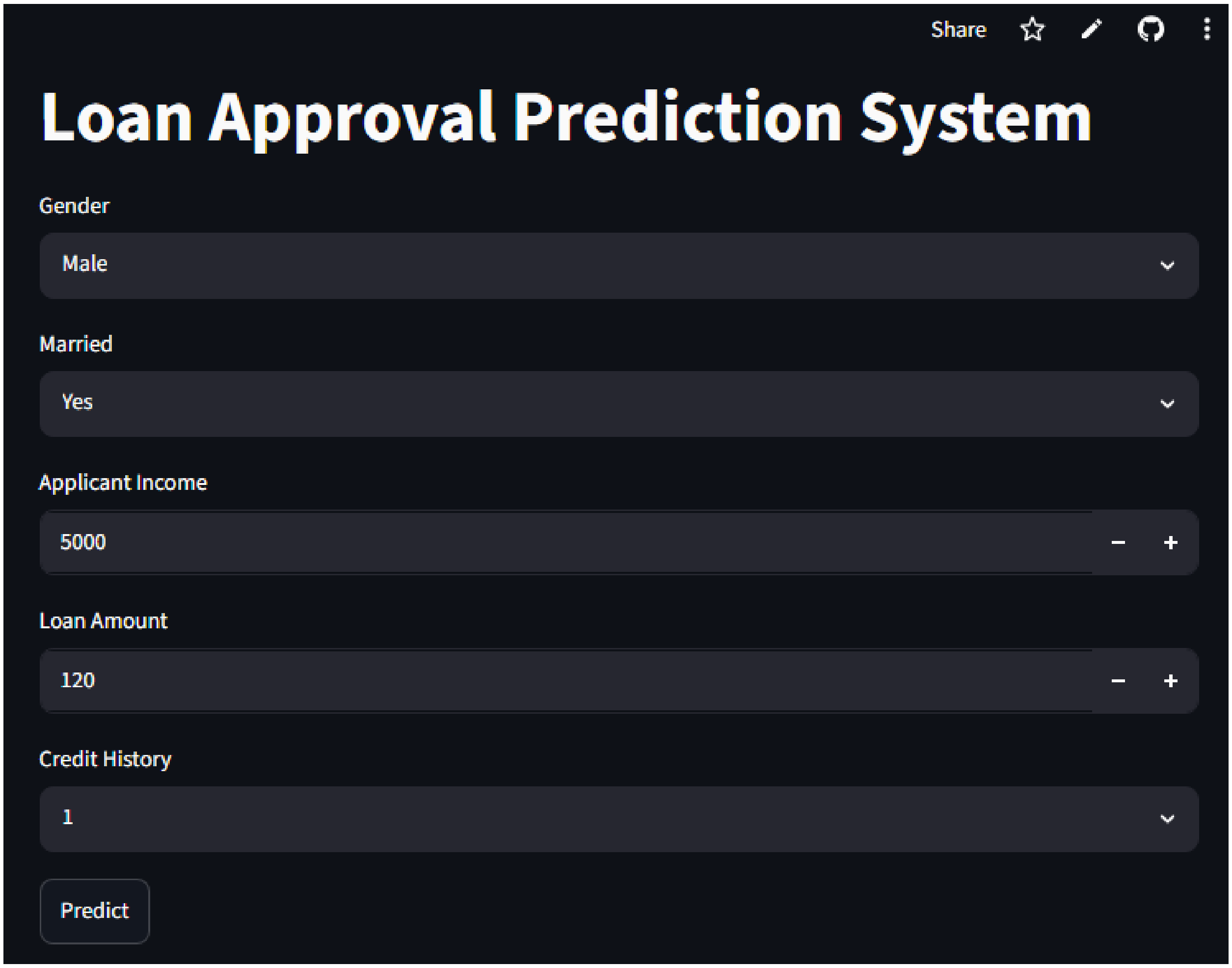



Figure 4.8: Loan Prediction Output !

# CHAPTER 5: RESULTS AND DISCUSSION

## 5.1Model Evaluation

After training the machine learning models, their performance was evaluated using the testing dataset. The models were assessed based on their prediction accuracy and classification performance.

The evaluation process helped in determining the most suitable model for loan approval prediction.

## 5.2Performance Metrics

The primary metric used for evaluation was Accuracy Score.

Accuracy is calculated as:

Accuracy = (Number of Correct Predictions / Total Predictions) × 100 A

higher accuracy value indicates better model performance.

## 5.3Results Obtained

The following machine learning models were trained and tested:

1. Logistic Regression
2. Random Forest Classifier

The obtained accuracy scores are shown below:

| Model | Accura y |
|---|---|
| Logistic Regression | 78.8% |
| Random Forest Classifier | 77.2% |

The results indicate that Logistic Regression achieved slightly better performance than Random Forest on the given dataset.

## 5.4Comparative Analysis

A comparison of both models shows that Logistic Regression provided higher prediction accuracy while maintaining simplicity and interpretability.

Although Random Forest is a powerful ensemble learning technique, its performance on this dataset was slightly lower than Logistic Regression.

Therefore, Logistic Regression was selected as the final model for deployment.

## 5.5 Discussion

The results demonstrate that machine learning techniques can effectively predict loan approval decisions using applicant information.

Among all available features, Credit History, Applicant Income, Loan Amount, and Total Income were observed to have a significant influence on loan approval outcomes.

The developed system successfully classified loan applications and achieved satisfactory prediction performance.

## 5.6 Deployment Results

The selected Logistic Regression model was integrated into a Streamlit web application.

The application allows users to enter applicant details and receive real-time loan approval predictions.

The deployment process confirmed that the trained model can be used in practical applications for assisting loan approval decisions.

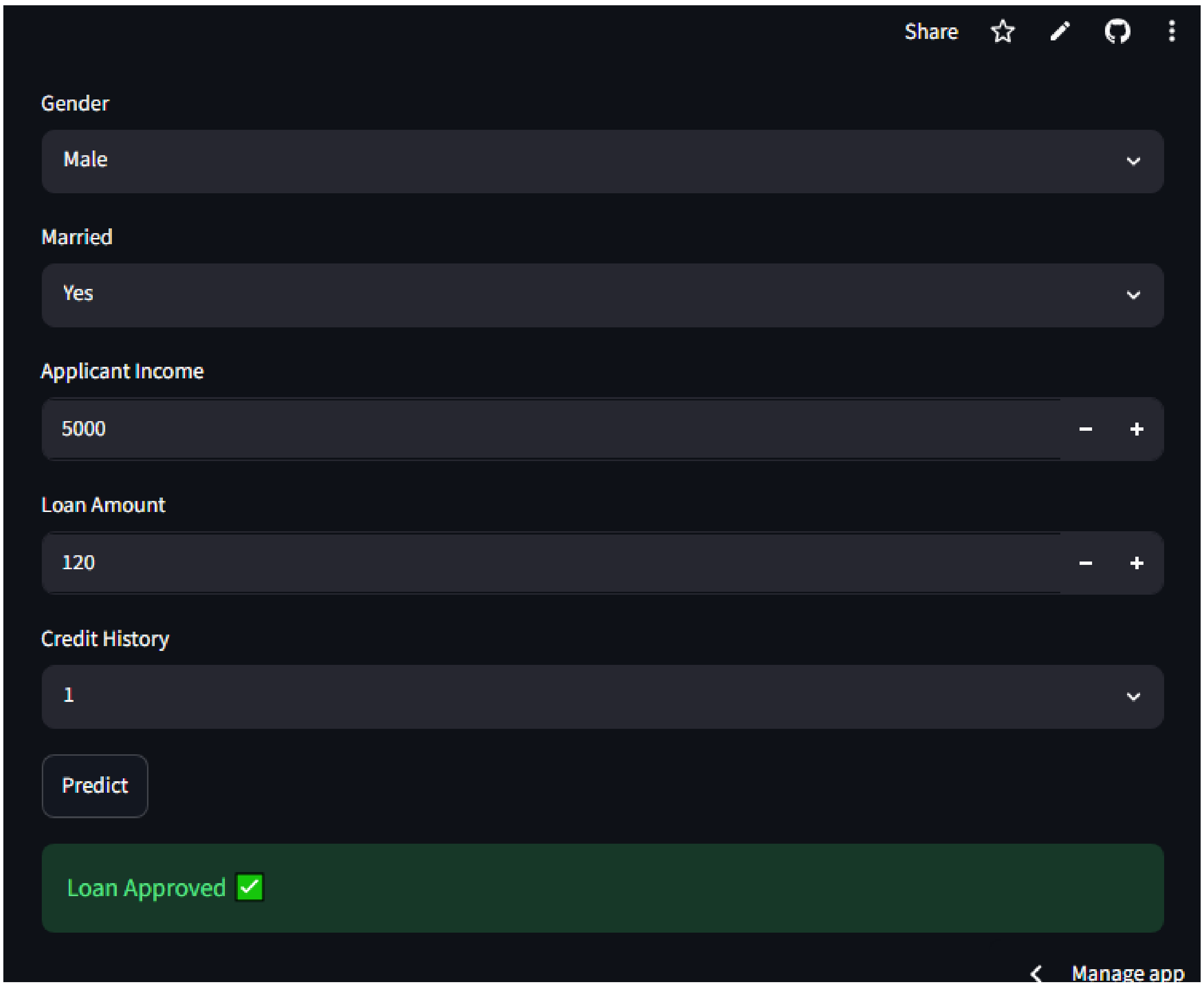
Model Accuracy Output

```
results = {
    "Model": ["Logistic Regression", "Random Forest"],
    "Accuracy": [accuracy, rf_accuracy]
}

pd.DataFrame(results)
```

| | Model | Accuracy |
|---|---|---|
| 0 | Logistic Regression | 0.788618 |
| 1 | Random Forest | 0.772358 |

Loan Prediction Result



# CHAPTER 6: CONCLUSION AND FUTURE SCOPE

## 6.1Conclusion

The Loan Approval Prediction System was successfully developed using Machine Learning techniques. The project involved various stages including data collection, data preprocessing, exploratory data analysis, feature engineering, model training, model evaluation, and deployment.

Two classification algorithms, Logistic Regression and Random Forest Classifier, were implemented and compared. After evaluation, Logistic Regression achieved an accuracy of 78.8%, which was higher than the accuracy obtained by Random Forest Classifier. Therefore, Logistic Regression was selected as the final model.

The selected model was deployed using Streamlit to create a user-friendly web application. The application enables users to enter loan applicant details and receive real-time loan approval predictions.

Overall, the project demonstrates how Machine Learning can be effectively used in the banking and financial sector to support faster, more accurate, and data-driven loan approval decisions.

## 6.2Future Scope

Although the developed system provides satisfactory results, several improvements can be made in the future.

- Larger and more diverse datasets can be used to improve prediction accuracy.
- Advanced machine learning algorithms such as XGBoost, Gradient Boosting, and Neural Networks can be implemented for better performance.
- Hyperparameter tuning techniques can be applied to optimize model accuracy.
- Additional applicant information can be incorporated to enhance prediction capability.
- The application can be integrated with real banking systems and databases for practical use.
- The Streamlit application can be enhanced with improved user interface design, dashboards, and data visualization features.

With these improvements, the system can become more robust, scalable, and suitable for real-world financial applications.

## REFERENCES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